Last-Iterate Rates of Stochastic Fast Krasnosel'skiĭ–Mann Iterations: Momentum, Implicit Variance Reduction, and Decaying Noise

Abstract

[Blue highlights the research questions and reading guide; red highlights the main answers and limitations.]

In this draft, we consider the problem of finding a fixed point of a nonexpansive operator in $\mathbb{R}^d$ from stochastic oracle evaluations. Many single-loop stochastic first-order methods, such as stochastic gradient descent and stochastic splitting methods, are stochastic Krasnosel'skiĭ–Mann (KM) iterations [3, 4]. For exact evaluations, the Fast KM method of Boţ and Nguyen [2] improves the last-iterate residual of KM from $o(1/\sqrt{k})$ to $o(1/k)$ through momentum. Under bounded variance, the last iterate of stochastic KM attains only $\log k/k^{1/6}$ [3]. We therefore study whether the momentum of Fast KM accelerates the last iterate when the operator values are noisy.

Key words. nonexpansive operator, fixed-point problem, Fast KM, stochastic oracle, almost-sure convergence.

1 Introduction

In this draft, we consider the problem of finding a fixed point of a nonexpansive operator $T : \mathbb{R}^d \to \mathbb{R}^d$ when only noisy evaluations of T are available. Many single-loop stochastic first-order methods are stochastic Krasnosel'skiĭ–Mann (KM) iterations of the form

$$x_{k+1} = (1 - \lambda_k)x_k + \lambda_k\big(Tx_k + e_k\big), \tag{1.1}$$

where e_k is the error of the kth oracle response [3–5]. Stochastic gradient descent, the stochastic proximal gradient method and stochastic backward–forward splitting are instances of this form; in the proximal gradient method, the error is in general conditionally biased.

For exact evaluations, the fixed-point residual of KM with a constant relaxation parameter satisfies $\|x_k - Tx_k\| = o(1/\sqrt{k})$ [2]. The Fast KM method of Boţ and Nguyen [2, Algorithm 2.1] improves this to

$$\|x_k - Tx_k\| = o(1/k), \qquad x_k \to x^\star \in \operatorname{Fix} T, \tag{1.2}$$

for a momentum parameter $\alpha > 2$.

With noisy evaluations, Tx_k in Fast KM is replaced by an oracle response $\widehat{T}_k(x_k) = Tx_k + e_k$. For the value at x_{k-1} there are two implementations.

- *Re-query with the same sample.* The sampled operator of iteration k is evaluated at x_k and at x_{k-1}. This uses two queries per iteration and an oracle that can evaluate one sample at two points [9, Assumption 2.3]. Variance-reduced methods build on this access [9, 11, 12].
- *Reuse.* The response $\widehat{T}_{k-1}(x_{k-1}) = Tx_{k-1} + e_{k-1}$ of the preceding step is stored and used again. This keeps one query per iteration, as in the deterministic method, and needs only fresh samples.

We thus study the reuse implementation. Table 1 raises three questions:

1. *(Q1) Last iterate of stochastic KM.* Under bounded variance, the best available last-iterate bound is $O(\log k/k^{1/6})$, whereas a randomly selected iterate attains $O(k^{-1/4})$. Does the last iterate necessarily lose this rate? (Section 3)

Table 1: Comparison of last-iterate rates $\mathbb{E}\|x_k - Tx_k\|$ for a nonexpansive T on $\mathbb{R}^d$.

Method	Momentum coefficient	Last-iterate rate
KM [2]	—	$o(1/\sqrt{k})$
Stochastic KM [3]	—	$O(\log k/k^{1/6})$
Fast KM [2]	$\frac{k}{k+\alpha} \to 1$	$o(1/k)$
Stochastic iKM [13]	$\Theta(1)$	$O\big(\frac{D}{k} + \frac{\sigma \log k}{\sqrt{k}}\big)$
Stochastic Fast KM, reuse (this paper)	$\frac{k}{k+\alpha} \to 1$	need not vanish (Prop. 4.2)
Same, $\mathbb{E}\sigma_k^2 \asymp k^{-q}$ (this paper)	$\frac{k}{k+\alpha} \to 1$	$o(1/k)$ for $q >$? (lower bound: $q > 1$)

One oracle query per iteration. Bounded variance in all stochastic rows except the last, which assumes decaying variance; the row with Prop. 4.2 also assumes $\mathbb{E}\sigma_k^2 \geq \sigma^2 > 0$. The lower bound refers to $\Omega(\sigma/\sqrt{N})$ after N samples [7, Theorem 2].

2. *(Q2) Implicit variance reduction in Fast KM.* Can the momentum of Fast KM reduce the variance implicitly and improve $O(\log k/k^{1/6})$ under bounded variance? (Section 4)

3. *(Q3) Fast KM under decaying noise.* Under bounded variance, the residual after k samples is $\Omega(\sigma/\sqrt{k})$ on the whole problem class [7, Theorem 2]. How fast must the variance decay for the reuse implementation to attain $o(1/k)$? (Section 5)

Notation. Throughout, $\mathbb{R}^d$ carries the Euclidean inner product $\langle\cdot,\cdot\rangle$, the associated norm $\|\cdot\|$ and the Borel sigma-algebra $\mathcal{B}$, and Id denotes the identity operator on $\mathbb{R}^d$. For positive sequences, $a_k \asymp b_k$ means that a_k/b_k is bounded from above and bounded away; $O(\cdot)$ and $o(\cdot)$ refer to $k \to \infty$. Equalities and inequalities between random variables hold $\mathbb{P}$-almost surely unless stated otherwise.

2 Preliminaries

The problem is

$$\text{find } x^\star \in \mathbb{R}^d \quad \text{such that} \quad Tx^\star = x^\star, \tag{FP}$$

where T satisfies the following assumption.

Assumption 2.1 (Operator)**.** $T : \mathbb{R}^d \to \mathbb{R}^d$ is nonexpansive, that is, $\|Tx - Ty\| \leq \|x - y\|$ for all $x, y \in \mathbb{R}^d$, and $\operatorname{Fix} T := \{x \in \mathbb{R}^d : Tx = x\}$ is nonempty.

The iterate accuracy is measured by the residual $\|x - Tx\|$. With $V := \mathrm{Id} - T$, nonexpansiveness of T is equivalent to

$$\langle Vx - Vy, x - y\rangle \geq \frac{1}{2}\|Vx - Vy\|^2, \qquad x, y \in \mathbb{R}^d. \tag{2.1}$$

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(\mathcal{F}_k)_{k\geq 0}$ a filtration, where $\mathcal{F}_k$ contains the information available before query k. At an $\mathcal{F}_k$-measurable point x_k, the oracle returns

$$\widehat{T}_k(x_k) = Tx_k + e_k \tag{2.2}$$

with an $\mathcal{F}_{k+1}$-measurable error e_k.

Assumption 2.2 (Oracle)**.** For every $k \geq 0$,

$$\mathbb{E}[e_k \mid \mathcal{F}_k] = 0 \qquad \text{and} \qquad \sigma_k^2 := \mathbb{E}\big[\|e_k\|^2 \;\big|\; \mathcal{F}_k\big] \text{ with } \mathbb{E}\sigma_k^2 < \infty. \tag{H1}$$

3 The last iterate of stochastic KM

In this section, we review existing convergence results for stochastic Krasnosel'skiĭ–Mann iterations.

For Problem (FP), the Krasnosel'skiĭ–Mann (KM) iteration reads

$$x_{k+1} = (1-\lambda_{k+1})x_k + \lambda_{k+1}Tx_k, \qquad k \geq 0, \tag{3.1}$$

with $x_0 \in \mathbb{R}^d$ and parameters $\lambda_k \in (0,1)$.

With a constant $\lambda_k \equiv \lambda$, the iterates converge to a point of $\operatorname{Fix} T$, and the rate is

$$\|x_k - Tx_k\| = o(1/\sqrt{k}).$$

The stochastic KM iteration is

$$x_{k+1} = (1-\lambda_{k+1})x_k + \lambda_{k+1}\widehat{T}_k(x_k), \qquad \widehat{T}_k(x_k) = Tx_k + e_k. \tag{3.2}$$

under bounded variance, a randomly selected iterate generated by (3.2) attains the order $k^{-1/4}$ [3, Corollary 5.4], [5]. This naturally introduce the following question:

(Q1) For a general nonexpansive T and bounded variance, can the last iterate of stochastic KM attain the order $k^{-1/4}$?

We recall the last-iterate bound of [3]. Let $D := \operatorname{dist}(x_0, \operatorname{Fix} T)$ and set $\tau_k := \sum_{j=1}^{k} \lambda_j(1-\lambda_j)$, $\varphi(y) := \min\{1, 1/\sqrt{\pi y}\}$ and

$$\nu_k^2 := \sum_{j=1}^{k} \big(w_j^k\big)^2 \, \mathbb{E}\sigma_{j-1}^2, \qquad w_j^k := \lambda_j \prod_{i=j+1}^{k} (1-\lambda_i),$$

so that τ_k is the effective time of the iteration and ν_k the scale of the averaged noise.

Assumption 3.1. Either T has bounded range and $\bar{\kappa} \geq \sup_x \|Tx - x_0\|$, or $S := \sum_{j\geq 2} \lambda_j \nu_{j-1} < \infty$ and $\bar{\kappa} \geq 2D + S$.

Theorem 3.1 (Last-iterate residual of stochastic KM [3, Theorem 2.10])**.** *Let Assumptions 2.1, 2.2 and 3.1 hold. Then, for every $k \geq 1$,*

$$\mathbb{E}\|x_k - Tx_k\| \leq \bar{\kappa}\,\varphi(\tau_k) + 2\sum_{j=2}^{k} \lambda_j \, \varphi(\tau_k - \tau_j)\, \nu_{j-1} + 4\nu_k. \tag{3.3}$$

We now consider the special case of Assumption 2.2 in which the second moments are uniformly bounded.

Assumption 3.2 (Bounded variance)**.** There exists a finite constant $\sigma \geq 0$ such that $\sup_{k\geq 0} \mathbb{E}\sigma_k^2 \leq \sigma^2$.

Corollary 3.2. *[3, Theorem 4.4] Under the assumptions of Theorem 3.1 and Assumption 3.2, let $\lambda_k = (k+1)^{-a}$ with $a \in (\frac{1}{2}, 1)$. Then $\tau_k \asymp k^{1-a}$ and it holds that*

$$\mathbb{E}\|x_k - Tx_k\| \lesssim \underbrace{\frac{1}{\sqrt{\tau_k}}}_{\text{deterministic}} + \underbrace{\sigma\sqrt{k}\,\lambda_k}_{\text{noise}} = \frac{1}{k^{(1-a)/2}} + \frac{\sigma}{k^{a-1/2}}.$$

Moreover, it holds that $\mathbb{E}\|x_k - Tx_k\| = O(\log k / k^{1/6})$ with $a = \frac{2}{3}$.

Answer to (Q1). In the literature examined here, No: for a general nonexpansive T,

$$\mathbb{E}\|x_k - Tx_k\| = O\Big(\frac{\log k}{k^{1/6}}\Big) \quad \text{(last iterate)}, \qquad \mathbb{E}\|x_R - Tx_R\| = O\Big(\frac{1}{k^{1/4}}\Big) \quad \text{(random iterate)}.$$

The limitations are

1. For exact evaluations,

$$\|x_{k+1} - Tx_{k+1}\| \leq \|x_k - Tx_k\| \quad \Longrightarrow \quad \|x_k - Tx_k\|^2 \leq \frac{D^2}{\tau_{k+1}};$$

   with noise, $\mathbb{E}\|x_k - Tx_k\|^2$ need not be monotone, and the summed bound reaches only x_R [3, Remark 5.3].

2. *Order of the last iterate:* in (3.3), $1/\sqrt{\tau_k} \asymp \sqrt{k}\,\lambda_k \asymp k^{-1/6}$ at $a = \frac{2}{3}$ (Corollary 3.2).

4 Implicit variance reduction in Fast KM

4.1 Fast KM iteration

Boţ and Nguyen [2] accelerate the KM iteration using Nesterov's momentum updates. For fixed parameters $\alpha > 2$ and $0 < s \le 1$, and initial points $x_0, x_1 \in \mathbb{R}^d$, their Fast KM method is

$$\begin{aligned} x_{k+1} = &\left(1 - \frac{s\alpha}{2(k+\alpha)}\right) x_k + \frac{(1-s)k}{k+\alpha}(x_k - x_{k-1}) + \frac{s\alpha}{2(k+\alpha)} Tx_k \\ &+ \frac{sk}{k+\alpha}(Tx_k - Tx_{k-1}), \qquad k \ge 1. \end{aligned} \tag{4.1}$$

The stepsize s can be absorbed into the operator: (4.1) applied to T coincides with (4.1) with $s = 1$ applied to $M := (1-s)\,\mathrm{Id} + sT$, which is nonexpansive with $\operatorname{Fix} M = \operatorname{Fix} T$, and an oracle for T with error e_k yields an oracle for M with error se_k. We therefore take $s = 1$ without loss of generality:

$$x_{k+1} = \left(1 - \frac{\alpha}{2(k+\alpha)}\right) x_k + \frac{\alpha}{2(k+\alpha)} Tx_k + \frac{k}{k+\alpha}(Tx_k - Tx_{k-1}). \tag{4.2}$$

4.2 The reuse implementation of Stochastic Fast KM

With noisy evaluations, Tx_k in (4.2) is replaced by the response $\widehat{T}_k(x_k)$, and Tx_{k-1} by the stored response $\widehat{T}_{k-1}(x_{k-1})$, so that each iteration queries the oracle once. Set

$$a_k := \frac{k + \alpha/2}{k+\alpha}, \qquad d_k := \frac{k}{k+\alpha}. \tag{4.3}$$

Algorithm 1 Stochastic Fast KM with reuse

Input: $x_0, x_1 \in \mathbb{R}^d$, $\alpha > 2$.
Query $\widehat{T}_0(x_0)$ and store it.
for $k = 1, 2, \ldots$ **do**
  Query $\widehat{T}_k(x_k) = Tx_k + e_k$.
  $x_{k+1} = (1 - a_k + d_k)\, x_k + (a_k - d_k)\, \widehat{T}_k(x_k) + d_k\big(\widehat{T}_k(x_k) - \widehat{T}_{k-1}(x_{k-1})\big)$.
  Store $\widehat{T}_k(x_k)$.
end for

Under bounded variance, no method guarantees a residual smaller than $D/K + \sigma/\sqrt{K}$ after K queries on the whole problem class [10, Theorem 4.6], [7, Theorem 2], whereas the last iterate of stochastic KM attains only $O(\log k / k^{1/6})$ (Section 3). In stochastic optimization, momentum averages the sampled gradients and reduces the variance implicitly [6, 8]. Algorithm 1 is stochastic KM with momentum and one query per iteration, which leads to the second question:

(Q2) Can the momentum reduce the variance implicitly and accelerate the last iterate of stochastic KM under bounded variance?

4.3 Noise propagation under reuse

Substituting (2.2) into the difference of oracle responses yields

$$\widehat{T}_k(x_k) - \widehat{T}_{k-1}(x_{k-1}) = Tx_k - Tx_{k-1} + e_k - e_{k-1}. \tag{4.4}$$

Relative to the exact update (4.2) evaluated at the current iterates, the total additive error is

$$\eta_k = (a_k - d_k)e_k + d_k(e_k - e_{k-1}) = a_k e_k - d_k e_{k-1}. \tag{4.5}$$

Since e_{k-1} is $\mathcal{F}_k$-measurable, Assumption 2.2 gives

$$\mathbb{E}[\eta_k \mid \mathcal{F}_k] = -d_k e_{k-1}. \tag{4.6}$$

Since $1 + d_k = 2a_k$, the update of Algorithm 1 reads $x_{k+1} = a_k x_k + a_k \widehat{T}_k(x_k) - d_k \widehat{T}_{k-1}(x_{k-1})$. The following lemma isolates the noise that this update propagates.

Lemma 4.1 (Noise propagation under reuse)**.** *Let Assumption 2.2 hold and set* $b_k := a_k a_{k-1} - d_k = \frac{\alpha(\alpha-2)}{4(k+\alpha)(k+\alpha-1)}$. *Define* $u_1 := \frac{1}{2} e_0$ *and*

$$u_{k+1} = a_k\, u_k + a_k\, e_k - d_k\, e_{k-1}, \qquad k \geq 1. \tag{4.7}$$

Then $z_k := x_k - u_k$ *satisfies the deterministic Fast KM recursion with errors* $\delta_k := Tx_k - Tz_k$, $\|\delta_k\| \leq \|u_k\|$. *Moreover, with* $v_k := u_k - a_{k-1} e_{k-1}$,

$$u_{k+1} = a_k\, e_k + v_{k+1}, \qquad v_{k+1} = a_k\, v_k + b_k\, e_{k-1}, \qquad v_1 = 0, \tag{4.8}$$

and

$$\mathbb{E}\|u_{k+1}\|^2 = a_k^2\, \mathbb{E}\sigma_k^2 + \mathbb{E}\|v_{k+1}\|^2, \qquad \mathbb{E}\|v_{k+1}\|^2 = a_k^2\, \mathbb{E}\|v_k\|^2 + b_k^2\, \mathbb{E}\sigma_{k-1}^2. \tag{4.9}$$

Proof. Subtracting (4.7) from the update gives $z_{k+1} = a_k z_k + a_k(Tz_k + \delta_k) - d_k(Tz_{k-1} + \delta_{k-1})$, and $\|\delta_k\| \leq \|x_k - z_k\|$ by nonexpansiveness. Substituting $u_k = v_k + a_{k-1} e_{k-1}$ into (4.7) gives (4.8). The vector v_k depends on $e_0, \ldots, e_{k-2}$ and is $\mathcal{F}_{k-1}$-measurable, and v_{k+1} is $\mathcal{F}_k$-measurable; conditional unbiasedness removes the cross terms in (4.9). □

In (4.9), the stored error e_{k-1} enters with the net coefficient $b_k = O(1/k^2)$: its two occurrences in consecutive updates cancel to leading order. The new error e_k enters with the weight $a_k \to 1$. Compared with momentum-based variance reduction, where the new noise enters with a tunable weight ρ_k [6, 8], the reuse implementation with fixed α reduces only the stored noise.

Proposition 4.2 (Parameters of Fast KM)**.** *Let* $T = 0$ *and assume* $\mathbb{E}\sigma_k^2 \geq \sigma^2 > 0$ *for all* k. *Then the iterates of Algorithm 1 satisfy* $\mathbb{E}\|x_{k+1} - Tx_{k+1}\|^2 \geq \sigma^2/4$ *for every* $k \geq 1$.

Proof. For $T = 0$, the update is $x_{k+1} = a_k x_k + a_k e_k - d_k e_{k-1}$. Since x_k and e_{k-1} are $\mathcal{F}_k$-measurable, conditional unbiasedness gives $\mathbb{E}\|x_{k+1}\|^2 = \mathbb{E}\|a_k x_k - d_k e_{k-1}\|^2 + a_k^2 \mathbb{E}\sigma_k^2 \geq \sigma^2/4$, because $a_k \geq \frac{1}{2}$. □

4.4 Horizon-dependent parameters

With parameters that vary with k, momentum does accelerate the last iterate. Consider the inertial KM iteration

$$y_k = x_k + \alpha_k(x_k - x_{k-1}), \quad z_k = x_k + \beta_k(x_k - x_{k-1}), \quad x_{k+1} = (1 - \lambda_k) y_k + \lambda_k \widehat{T}_k(z_k), \tag{4.10}$$

which contains Fast KM as a particular choice of parameters [13, Appendix A]. For $K \geq 2$ and $0 < a \leq \frac{K \log 2}{\log K} - 1$, set $\omega_k := \log\bigl(1 + K^{a-(a+1)k/K}\bigr)$, $r_k := 2^{\omega_{k-1}/\omega_k} - 1$, $u_k' := \frac{3 r_k}{2(1+r_k)(2+r_k)}$, and

$$\beta_1 = 0, \quad \beta_k = \frac{4}{3 r_{k-1}}, \qquad \lambda_k = \frac{K r_k\, \omega_{k-1}}{4(1 + 1/a)\, \omega_0}\, u_k', \qquad \alpha_k = \frac{u_k' - \lambda_k}{1 - \lambda_k}\, \beta_k. \tag{4.11}$$

Theorem 4.3 (Varying parameters [13, Theorem 1])**.** *Let Assumption 2.1 hold, let the errors be conditionally unbiased with* $\mathbb{E}[\|e_k\|^2 \mid \mathcal{F}_k] \leq \sigma^2$, *and generate the iterates by* (4.10) *with* (4.11). *Then, for an absolute constant* $C > 0$,

$$\Bigl(\mathbb{E}\|x_K - Tx_K\|^2\Bigr)^{1/2} \leq C\Bigl(\frac{1 + 1/a}{K}\, D + (1 + a)\frac{\sigma \log K}{\sqrt{K}}\Bigr).$$

Corollary 4.4. *For* $a = \Theta(1)$, $\mathbb{E}\|x_K - Tx_K\| \lesssim \dfrac{D}{K} + \dfrac{\sigma \log K}{\sqrt{K}}$.

Remark 4.5 (Answer to (Q2)).

1. *Mechanism.* Momentum reduces the variance implicitly: reuse cancels the stored noise (Lemma 4.1), and parameters varying with k lower the weight of the new noise (Theorem 4.3).

2. *Parameters must scale with the horizon.* With the parameters of Fast KM, the new noise keeps the weight $a_k \to 1$, and the residual does not vanish (Proposition 4.2). The available acceleration uses parameters that depend on the horizon K and bounds the iterate at step K only.

3. *Trade-off.* For exact evaluations, Fast KM improves KM uniformly. Under bounded variance, the parameter a trades the deterministic factor $1 + 1/a$ against the noise factor $1 + a$, and the noise term $\sigma/\sqrt{K}$ dominates D/K.

5 Fast KM under decaying noise

Under bounded variance, no method guarantees a residual smaller than $\sigma/\sqrt{N}$ after N queries [7, Theorem 2]. Since Corollary 4.4 is already within a logarithmic factor of this order, the rate $o(1/k)$ requires $\sigma/\sqrt{N_k} = o(1/k)$, that is, $N_k \gg k^2$ samples in the first k steps. The noise itself has to decay.

(Q3) How fast must the variance σ_k^2 decay for the reuse implementation of stochastic Fast KM to attain $\mathbb{E}\|x_k - Tx_k\| = o(1/k)$?

With mini-batches $m_k \asymp k^q$, the lower bound requires only $q > 1$.
For the almost-sure counterpart, the objective is to identify noise conditions under which

$$k\|x_k - Tx_k\| \longrightarrow 0 \qquad \text{almost surely,} \tag{5.1}$$

and, for a Fix$T$-valued random variable x_∞,

$$x_k \to x_\infty \qquad \text{almost surely.} \tag{5.2}$$

The candidate condition is the decaying-noise condition of [1, Theorem 3.1]: $\sum_k s_k^2 t_k^2 \sigma_k^2 < \infty$ or $\sum_k s_k t_k \sigma_k < \infty$ almost surely, where s_k is the stepsize and $t_k := 1 + \sum_{i \geq k} \prod_{j=k}^{i} \alpha_j$ for the extrapolation coefficients α_k. For Algorithm 1, $s_k \equiv 1$ and $\alpha_k = d_k$, so $t_k = \frac{k+\alpha-1}{\alpha-1}$ and the condition reads

$$\text{(N6-}\ell^2\text{)} \ \sum_k (k+1)^2 \sigma_k^2 < \infty \qquad \text{or} \qquad \text{(N6-}\ell^1\text{)} \ \sum_k (k+1)\sigma_k < \infty \tag{5.3}$$

almost surely. With mini-batches $m_k \asymp k^q$ and $\sigma_k^2 = \sigma^2/m_k$, (N6-$\ell^2$) holds if and only if $q > 3$, and (N6-ℓ^1) if and only if $q > 4$.

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